

assignment1 REview basic Math

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:24pm

Quiz Instructions

2 attempts/

Refresh your basic Math skills !

There is a tutorial in the dropbox for most of the questions but TRY first

The latest attempt counts.

Also: you have 2 tutorials in dropbox about conversions and scientific notation if you need refresher



Question 1 4.4 pts

Convert 1.2×10^{-3} to decimal notation.

☐

1.200

☐

0.00012

☐

0.0120

☐

0.1200

☐

0.0012



Question 2 4.4 pts

The shortest wavelength of visible light is approximately 400 nm. Express this wavelength in centimeters.

☐

4×10^{-9} cm

☐

400×10^{-11} cm

☐

4×10^{-5} cm



$$4 \times 10^{-11} \text{ cm}$$



$$4 \times 10^{-7} \text{ cm}$$



Question 3 4.4 pts

The following conversion equivalents are given:

$$1 \text{ gal} = 231 \text{ in}^3 \quad 1 \text{ ft} = 12 \text{ in} \quad 1 \text{ min} = 60 \text{ s}$$

A pipe delivers water at the rate of 95 gal/min. The rate in ft^3/s , is closest to:



$$0.15$$



$$0.19$$



$$0.14$$



$$0.21$$



$$0.17$$



Question 4 4.4 pts

0.0001776 can also be expressed as



$$17.72 \times 10^4.$$



$$177.2 \times 10^7.$$



$$1772 \times 10^5.$$



$$1.776 \times 10^{-4}.$$



$$1.776 \times 10^{-3}.$$



Question 5 4.4 pts

$0.00325 \times 10^{-8} \text{ cm}$ can also be expressed in mm as

☐ $3.25 \times 10^{-8} \text{ mm.}$ ☐ $3.25 \times 10^{-11} \text{ mm.}$ ☐ $3.25 \times 10^{-10} \text{ mm.}$ ☐ $3.25 \times 10^{-12} \text{ mm.}$ ☐ $3.25 \times 10^{-9} \text{ mm.}$ 

Question 6 4.4 pts

Express $(4.3 \times 10^6)^{-1/2}$ in scientific notation.

☐ 2.1×10^{-5} ☐ 2.1×10^3 ☐ 4.8×10^{-4} ☐ 2.1×10^4 

Question 7 4.4 pts

A weight lifter can bench press 171kg How many milligrams (mg) is this?

☐ $1.71 \times 10^8 \text{ mg}$ ☐ $1.71 \times 10^6 \text{ mg}$ ☐ $1.71 \times 10^9 \text{ mg}$ ☐ $1.71 \times 10^7 \text{ mg}$ 

Question 8 4.4 pts

The wavelength of a certain laser is 0.35 micrometers (0.35 μm) where 1 μm is a millionth of a meter. Express this wavelength in nanometers.

☐ $3.5 \times 10^1 \text{ nm}$



$3.5 \times 10^3 \text{ nm}$



$3.5 \times 10^2 \text{ nm}$



$3.5 \times 10^4 \text{ nm}$



Question 9 4.4 pts

Convert 1.2×10^{-3} to decimal notation.



0.0012



1.200



0.1200



0.00012



0.0120



Question 10 4.4 pts

In addition to $1 \text{ m} = 39.37 \text{ in.}$, the following exact conversion equivalents are given below and $1 \text{ min} = 60 \text{ s}$. If a particle has a velocity of 8.4 miles per hour, its velocity, in m/s, is closest to

$1 \text{ mile} = 5280 \text{ ft}$

$1 \text{ ft} = 12 \text{ in}$

$1 \text{ hour} = 60 \text{ min}$



3.4 m/s.



4.5 m/s.



3.0 m/s.



3.8 m/s.



4.1 m/s.



Question 11 4.4 pts

A speed of 65 miles per hour is the same as

☐

32 m/s

☐

29 m/s

☐

42 m/s

☐

24 m/s

☐

37 m/s



Question 12 4.4 pts

0.0001776 can also be expressed as

☐

1772×10^5 .

☐

17.72×10^4 .

☐

177.2×10^7 .

☐

1.776×10^{-3} .

☐

1.776×10^{-4} .



Question 13 4.4 pts

0.00325×10^{-8} cm can also be expressed in mm as

☐

3.25×10^{-9} mm.

☐

3.25×10^{-11} mm.

☐

3.25×10^{-8} mm.

☐

3.25×10^{-12} mm.

☐

3.25×10^{-10} mm.



Question 14 4.4 pts

Convert a speed of 4.50 km/h to units of ft/min. (1.00 m = 3.28 ft)

☐

246 ft/min

☐

82.3 ft/min

☐

165 ft/min

☐

886 ft/min

☐

0.246 ft/min

☐

Question 15 4.4 pts

The following conversion equivalents are given:

$$1 \text{ m} = 100 \text{ cm} \quad 1 \text{ in} = 2.54 \text{ cm} \quad 1 \text{ ft} = 12 \text{ in}$$

A bin has a volume of 1.5 m^3 . The volume of the bin, in ft^3 , is closest to:

☐

59

☐

53

☐

35

☐

47

☐

41

☐

Question 16 4.4 pts

One light-year is the distance light travels in one year. The speed of light is about 300,000 km/s (3×10^5 km/s). How far is 1 light-year?

☐
 $1.08 \times 10^9 \text{ km}$
☐
 $9.46 \times 10^{12} \text{ km}$
☐
 $1.8 \times 10^7 \text{ km}$
☐
 $3 \times 10^5 \text{ km}$

**Question 17 4.4 pts**

Light travels at a speed of 300,000 km/s. About how far is a light-year?

☐

10 million meters

☐

10 trillion km

☐

10 million km

☐

10 billion km

☐

300,000 km

**Question 18 4.4 pts**

Suppose you know the speed of a spacecraft in kilometers per second. How would you calculate its speed in kilometers per hour?

☐

multiply by 3600

☐

Multiply by 24.

☐

Multiply by 60 and then divide by 60 again.

☐

Divide by 24.

**Question 19 4.4 pts**

How many seconds are in one year? (Calculate this, do not look it up.)

☐

about 3600 (3.6×10^3)

☐

about 380 million (380,000,000, or 3.8×10^8)

☐

about 86 thousand (86,000, or 8.6×10^5)

☐

about 30 million (30,000,000, or 3×10^7)

**Question 20 4.4 pts**

One light-hour is the distance light travels in one hour. The speed of light is about 300,000 km/s (3×10^5 km/s). If Jupiter is 0.72 light hours from the Sun, how far is this?

☐

1.5 billion km (1.5×10^9 km)

☐

13 million km (1.3×10^7 km)

☐

216 thousand km (2.16×10^5 km)

☐

778 million km (7.78×10^8 km)



Question 21 4.4 pts

The planet Mars is, on average, about 228 million km from the Sun. How long does it take light from the Sun to reach Mars? (Recall that the speed of light is about 300,000 km/s.)

☐

about 1.52 hours

☐

about 12.7 minutes

☐

about 1.52 light seconds

☐

about 8.4 minutes



Question 22 4.4 pts

Our solar system is located about 27,000 light-years from the galactic center. About how far does our solar system travel in one orbit?

hint: distance around a circle is $2\pi R$

☐

170,000 light-years

☐

85,000 light-years

☐

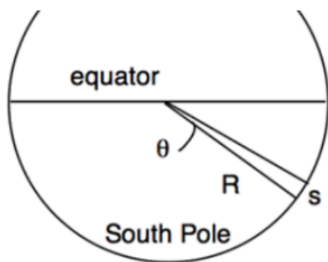
100,000 light-years

☐

54,000 light-years



Question 23 4.4 pts



The sketch is a cross section through the earth, showing two points that have a difference in latitude of angle θ . The mean radius R of the earth is 6.371×10^6 meters. A nautical mile is defined as 1852 meters exactly. Using these data, calculate how many nautical miles there are between two points on the earth's surface that have the same longitude but have latitudes that differ by 1.000 degree. (Just in case: the definition of angle, in radians, is the arc of a circle it subtends divided by the radius of the circle. A circle is 2π radians or 360° .)

There are = _____ nautical miles between two points with latitudes differing by 1.000° . (Enter your answer as a number, without exponents and without units, but with appropriate significant figures.)

☐

100 miles

☐

10 miles

☐

60 miles

☐

10^5 miles

Not saved

Submit Quiz